

220B Discussion 5

Problem 1: Stein vs. UMVUE

Let $\mathbf{X} \sim \mathcal{N}_p(\theta, I_p)$ with $p \geq 3$, under squared-error loss.

- (a) Recall the standard MSE decomposition of $R(\theta, \delta)$ into a sum of variances and squared biases. Write it out for an arbitrary estimator $\delta(\mathbf{X}) = (\delta_1(\mathbf{X}), \dots, \delta_p(\mathbf{X}))$.
- (b) Compute the coordinatewise bias $b_i(\theta) := \mathbb{E}_\theta[\delta_i^{JS}(\mathbf{X})] - \theta_i$ for the James-Stein estimator. Express it as a single expectation. Then evaluate $b_i(\mathbf{0})$ for every i . (You should find a clean answer.)
- (c) From part (b) and Lecture 10's risk formula, deduce the value of $\sum_{i=1}^p \text{Var}_0(\delta_i^{JS})$, i.e., the total variance of δ^{JS} at $\theta = \mathbf{0}$.
- (d) The MLE $\delta^0(\mathbf{X}) = \mathbf{X}$ is the UMVUE of θ . State the UMVUE optimality property precisely (one sentence). Now compare $\sum_i \text{Var}_0(X_i)$ to your answer from (c). Discuss with your group: is this a counterexample to UMVUE optimality? If not, exactly which hypothesis of UMVUE does δ^{JS} fail to satisfy, and where would you look in the data to verify the failure? Be prepared to defend your answer against a classmate who claims δ^{JS} is a valid UMVUE-comparable estimator at $\theta = \mathbf{0}$.

Problem 2: Lower MSE does not buy you better inference

Continue with $\mathbf{X} \sim \mathcal{N}_p(\theta, I_p)$, $p \geq 3$. Suppose a colleague proposes the following procedure for inference on θ_1 :

“Since δ_1^{JS} has lower MSE than X_1 , use $\delta_1^{JS} \pm 1.96$ as a 95% confidence interval for θ_1 .”

You suspect this is wrong. Let's diagnose what goes wrong.

(a) For $X_1 \sim \mathcal{N}(\theta_1, 1)$ (the marginal of $\mathbf{X}$ on coordinate 1), what is the coverage probability of the interval $X_1 \pm 1.96$ for θ_1 ? Why does it not depend on θ or on the other coordinates?

(b) Now consider the colleague's interval $\delta_1^{JS} \pm 1.96$. Write down its coverage probability as $\mathbb{P}_\theta(|\delta_1^{JS} - \theta_1| \leq 1.96)$ and explain in words why this *does* depend on θ (not just on θ_1).

(c) Show using a brief simulation argument (or careful reasoning about the JS sampling distribution) that for $\theta = (1, 1, \dots, 1)^T \in \mathbb{R}^{10}$, the coverage of the colleague's interval is *higher* than 0.95, even though its center is biased toward 0. Explain why this is not a refutation of your suspicion but rather a more subtle failure.

(d) Suppose instead the colleague uses the “correctly calibrated” interval $\delta_1^{JS} \pm 1.96 \cdot \text{SD}_\theta(\delta_1^{JS})$, where $\text{SD}_\theta(\delta_1^{JS})$ is the true sampling standard deviation of δ_1^{JS} (smaller than 1 because of shrinkage). What happens to the coverage now, and why? Tie your answer back to the bias-variance decomposition from Problem 1: which component of MSE is *invisible* to interval coverage, and which component is *lethal* to it?